The finite-energy 5D cross section integrated by parts in $M^2_{q\bar{q}}$

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Abstract

The finite-energy (FE) constrained DIS cross section with the dipole evolved to the rapidity of $x_{\mathbb{P}}$ is a five-dimensional integral whose $M^2_{q\bar{q}}$ integrand oscillates over the whole kinematic range. Integrating by parts in $M^2_{q\bar{q}}$ rewrites it exactly as the four-dimensional FE cross section with the dipole at $x_{\mathbb{P}} = 1$, plus a correction proportional to the rapidity derivative of the target amplitude. The correction vanishes wherever the dipole does not depend on rapidity, which for a BK solution starting at x_0 is most of the $M^2_{q\bar{q}}$ range. Numerically, this reduces the Monte Carlo error at a fixed number of points by a factor 12–17, and it makes adaptive cubature practical.

1 The cross section

We follow the implementation in `BK_FE_5D_evolve_in_xP.py`. For one quark flavour of mass m and polarization $\lambda = L, T$,

$$\sigma^{5\mathrm{D}}_\lambda = \int \mathrm{d}u\,\mathrm{d}u'\,\mathrm{d}\theta\,\mathrm{d}z \int_{4m^2}^{W^2} \mathrm{d}M^2_{q\bar{q}}\, P_\lambda(u,u',\theta,z)\, T\big(Y(M^2_{q\bar{q}})\big)\, \Theta\!\left(M^2_{q\bar{q}} - \frac{m^2}{z(1-z)}\right) \frac{z(1-z)}{4\pi}\, J_0(kr), \tag{1}$$

where u and u' are the dipole sizes in the amplitude and its conjugate, θ is the angle between them, and

$$r^2 = |\mathbf{u} - \mathbf{u}'|^2 = u^2 + u'^2 - 2uu'\cos\theta, \qquad k^2 = z(1-z)\,M^2_{q\bar{q}} - m^2, \tag{2}$$

$$W^2 = Q^2\left(\frac{1}{x_B} - 1\right), \qquad Y(M^2_{q\bar{q}}) = \ln\frac{x_0}{x_{\mathbb{P}}}, \qquad x_{\mathbb{P}} = \frac{M^2_{q\bar{q}} + Q^2}{W^2 + Q^2}. \tag{3}$$

The prefactor collects everything that does not depend on the target:

$$P_\lambda(u,u',\theta,z) = \frac{\sigma_0}{2}\,\frac{1}{4\pi}\,\frac{2\pi\,uu'}{z(1-z)}\,\big|\psi_\lambda(u,u',\theta,z)\big|^2, \tag{4}$$

with $|\psi_\lambda|^2$ the product of the light-cone wavefunction and its conjugate. The target amplitude

$$T(Y) = 1 - S_Y(u) - S_Y(u') + S_{4,Y}(u,u',\theta,z) \tag{5}$$

is built from the dipole S_Y and the quadrupole $S_{4,Y}$, all at rapidity Y.

The structure that matters below is that **only T depends on $M^2_{q\bar{q}}$**, through the rapidity $Y(M^2_{q\bar{q}})$, and that the phase-space kernel is $J_0(kr)$ with k^2 linear in $M^2_{q\bar{q}}$.

For comparison, the 4D cross section with the dipole evolved to x_B is

$$\sigma^{4\mathrm{D}}_\lambda[Y] = \int \mathrm{d}u\,\mathrm{d}u'\,\mathrm{d}\theta\,\mathrm{d}z\; P_\lambda(u,u',\theta,z)\, T(Y)\, \frac{\zeta J_1(\zeta)}{2\pi r^2}, \qquad \zeta = r\sqrt{z(1-z)\,M^2_{\max} - m^2}, \tag{6}$$

with $M^2_{\max} = Q^2(1-x_B)/x_B = W^2$ and the integrand set to zero where the square root is not real. `BK_FE_4D_evolve_in_xB.py` uses $Y = \ln(x_0/x_B)$; here the rapidity is left as an argument.

2 Integration by parts

Fix u, u', θ and z. The Θ function starts the $M^2_{q\bar{q}}$ integral at the threshold $M^2_{q\bar{q}} = m^2/[z(1-z)] \geq 4m^2$, where $k = 0$. The upper limit $M^2_{q\bar{q}} = W^2$ corresponds to $k = K$, with

$$K^2 = z(1-z)\,W^2 - m^2. \tag{7}$$

If $K^2 \leq 0$ the integrand vanishes, and there is nothing to do. Otherwise, change variables to $s = k^2$, so that $\mathrm{d}s = z(1-z)\,\mathrm{d}M^2_{q\bar{q}}$. The $M^2_{q\bar{q}}$ integral in (1) becomes

$$I \equiv \int_{4m^2}^{W^2} \mathrm{d}M^2_{q\bar{q}}\,\Theta(\ldots)\,\frac{z(1-z)}{4\pi}\,J_0(kr)\,T = \frac{1}{4\pi}\int_0^{K^2} \mathrm{d}s\;J_0(\sqrt{s}\,r)\,T. \tag{8}$$

The kernel has an elementary antiderivative in s. From $\frac{\mathrm{d}}{\mathrm{d}x}\big[xJ_1(x)\big] = xJ_0(x)$ it follows that

$$F(s) \equiv \frac{2\sqrt{s}\,J_1(\sqrt{s}\,r)}{r}, \qquad \frac{\mathrm{d}F}{\mathrm{d}s} = J_0(\sqrt{s}\,r), \qquad F(0) = 0. \tag{9}$$

Integrating by parts,

$$I = \frac{1}{4\pi}\Big[F(s)\,T\Big]_0^{K^2} - \frac{1}{4\pi}\int_0^{K^2} \mathrm{d}s\;F(s)\,\frac{\mathrm{d}T}{\mathrm{d}s}. \tag{10}$$

The lower boundary term vanishes because $F(0) = 0$. In the remaining integral, $\mathrm{d}s\,(\mathrm{d}T/\mathrm{d}s) = \mathrm{d}M^2_{q\bar{q}}\,(\mathrm{d}T/\mathrm{d}M^2_{q\bar{q}})$, and by the chain rule

$$\frac{\mathrm{d}T}{\mathrm{d}M^2_{q\bar{q}}} = \frac{\mathrm{d}T}{\mathrm{d}Y}\,\frac{\mathrm{d}Y}{\mathrm{d}M^2_{q\bar{q}}} = -\frac{1}{M^2_{q\bar{q}} + Q^2}\,\frac{\mathrm{d}T}{\mathrm{d}Y}. \tag{11}$$

Therefore

$$\boxed{I = \frac{KJ_1(Kr)}{2\pi r}\,T(Y_W) + \frac{1}{4\pi}\int \mathrm{d}M^2_{q\bar{q}}\,\frac{2kJ_1(kr)}{r}\,\frac{1}{M^2_{q\bar{q}} + Q^2}\,\frac{\mathrm{d}T}{\mathrm{d}Y}\bigg|_{Y(M^2_{q\bar{q}})}} \tag{12}$$

where the integral runs over the same range as before, from threshold to W^2, and $Y_W = Y(W^2) = \ln x_0$, the rapidity at $x_{\mathbb{P}} = 1$.

3 Result

The boundary term is the 4D cross section. With $\zeta = Kr$, the first term of (12) is

$$\frac{KJ_1(Kr)}{2\pi r} = \frac{\zeta J_1(\zeta)}{2\pi r^2}, \qquad \zeta = r\sqrt{z(1-z)\,W^2 - m^2}, \tag{13}$$

which is exactly the kernel of (6), because $M^2_{\max} = W^2$. Multiplying by P_λ and integrating over u, u', θ and z gives

$$\boxed{\sigma_\lambda^{5\mathrm{D}} = \sigma_\lambda^{4\mathrm{D}}\big[Y = \ln x_0\big] + \Delta\sigma_\lambda} \tag{14}$$

with the correction

$$\begin{aligned}\Delta\sigma_\lambda = \int \mathrm{d}u\,\mathrm{d}u'\,\mathrm{d}\theta\,\mathrm{d}z\,\mathrm{d}M^2_{q\bar{q}}\;P_\lambda(u,u',\theta,z)\;\Theta\!\left(M^2_{q\bar{q}} - \frac{m^2}{z(1-z)}\right)\\ \times\,\frac{1}{4\pi}\,\frac{2kJ_1(kr)}{r}\,\frac{1}{M^2_{q\bar{q}} + Q^2}\,\frac{\mathrm{d}T}{\mathrm{d}Y}\bigg|_{Y(M^2_{q\bar{q}})}.\end{aligned} \tag{15}$$

Interpretation. The 5D cross section equals the 4D one evaluated with the target at $x_{\mathbb{P}} = 1$, plus a term driven by how fast the evolution changes the target amplitude, $\partial T/\partial Y$, weighted by a smooth kernel. At $x_{\mathbb{P}} = 1$ we have $Y = \ln x_0 < 0$. A BK solution starting at x_0 is not evolved backwards, and the code evaluates the dipole at $Y < 0$ at its initial condition $Y = 0$. So the boundary term is the 4D cross section with the unevolved dipole.

The correction lives at small $M^2_{q\bar{q}}$. The derivative $\mathrm{d}T/\mathrm{d}Y$ vanishes wherever the dipole does not depend on Y. For a BK solution tabulated in $Y \in [Y_{\min}, Y_{\max}]$, with the dipole frozen outside, the integrand of (15) is non-zero only for

$$M^2(Y_{\max}) \le M^2_{q\bar{q}} \le M^2(Y_{\min}), \qquad M^2(Y) = x_0 \left(W^2 + Q^2\right) e^{-Y} - Q^2. \tag{16}$$

With $Y_{\min} = 0$, the upper end corresponds to $x_{\mathbb{P}} = x_0$. For $Q^2 = 2$ GeV^2, $x_B = 10^{-3}$ and $x_0 = 0.01$, the correction covers $M^2_{q\bar{q}} \lesssim 18$ GeV^2, while the original integral extends to $W^2 \approx 2000$ GeV^2.

Validity. Integration by parts requires T to be absolutely continuous in $M^2_{q\bar{q}}$, not smooth. Freezing the dipole at the edges of its rapidity range is continuous; $\mathrm{d}T/\mathrm{d}Y$ then jumps there, which is harmless. The kernel of the correction is regular at $r \to 0$, where $2kJ_1(kr)/r \to k^2$.

The derivation assumes that T is the only $M^2_{q\bar{q}}$-dependent factor and that the kernel is $J_0(kr)$ with k^2 linear in $M^2_{q\bar{q}}$. If either changes, for example through an $M^2_{q\bar{q}}$-dependent factor elsewhere, the derivation has to be redone.

4 Why this helps numerically

At fixed u, u', θ, z, the original integrand oscillates in $M^2_{q\bar{q}}$ through $J_0(kr)$ over the whole range up to W^2. Integrated over $M^2_{q\bar{q}}$, these oscillations produce the kernel $\zeta J_1(\zeta)/(2\pi r^2)$ of the 4D cross section. That kernel is sharply peaked at $r = 0$, with a width of about $1/\sqrt{z(1-z)W^2}$, roughly 0.05 GeV^{-1} in the example above. The peak emerges from a near-cancellation between large positive and negative contributions. A Monte Carlo estimate samples both and has a correspondingly large variance.

After the integration by parts, that cancellation is done analytically, in the antiderivative $F(s)$. What is left to integrate numerically is

- the 4D boundary term. Its peak at $r = 0$ is handled by integrating over $\mathbf{u}$ and the relative vector $\mathbf{r} = \mathbf{u} - \mathbf{u}'$ in polar coordinates, with $|\mathbf{r}|$ on a logarithmic scale, instead of over u, u' and θ;
- the 5D correction, over a short $M^2_{q\bar{q}}$ range where k is small and the kernel barely oscillates.

Table 1 shows the effect for $Q^2 = 2$ GeV^2, $x_B = 10^{-3}$, $m = 0.14$ GeV, $Z_f^2 = 2/3$, the MV-initial-condition BK solution of `data/mv.dat`, and the finite-N_c quadrupole. The numbers are structure functions $F_\lambda = Q^2\sigma_\lambda/(4\pi^2\alpha_{\mathrm{em}})$.

5 Implementation

In the C++ code (`cpp/`), the boundary term is `FE4DIntegrand` constructed with $Y = \ln x_0$, and the correction is `FE5DCorrectionIntegrand` in `src/integrands.hpp`. They are combined in `fe5d_structure_functions` in `src/cross_sections.hpp`. The option `--fe5d-method original` integrates the original form (1) instead. When the dipole depends on rapidity all the way to $x_{\mathbb{P}} = 1$ (as with `--large-x-power`), the correction (15) extends over the whole $M^2_{q\bar{q}}$

	F_L	F_T
Vegas, 10^6 points, original form	0.0752 ± 0.0020	0.2712 ± 0.0044
Vegas, 10^6 points, integrated by parts	0.07620 ± 0.00012	0.27574 ± 0.00038
of which the boundary term	0.06184 ± 0.00012	0.24603 ± 0.00038
of which the correction	0.014362 ± 0.000006	0.029712 ± 0.000011
Cuhre, $\epsilon_{\mathrm{rel}} = 10^{-4}$, integrated by parts	0.076391 ± 0.000006	0.276422 ± 0.000025
Python (vegas), mean of three runs of 10^7 points	0.0761 ± 0.0005	0.2769 ± 0.0011

Table 1: FE 5D structure functions. Both Vegas results use the same integration variables (relative coordinates, see above) and 10^6 points per iteration for each integral; they differ only in the integration by parts. Vegas errors are statistical. The Cuhre error is its own estimate, which covered the change to a ten times smaller ϵ_{rel}. The Python error is the scatter of the three runs, about twice the error vegas reports for one run.

range and the advantage of this form is largely lost; the code then integrates the $M^2_{q\bar{q}}$ integral at each point with a Filon rule instead (`--fe5d-method auto`, the default).

The derivative is a central finite difference,

$$\frac{\mathrm{d}T}{\mathrm{d}Y} \approx \frac{T(Y_+) - T(Y_-)}{Y_+ - Y_-}, \qquad Y_\pm = \mathrm{clamp}(Y \pm h,\ Y_{\min},\ Y_{\max}), \quad h = 10^{-4}, \tag{17}$$

taken as zero when $Y_+ \le Y_-$. It vanishes identically where the dipole is frozen, and it needs nothing from the dipole model except its values and its rapidity range, so it works with any interpolation of the BK solution. Its error is about 10^{-8} relative.

A unit test (`tests/test_sampling.cpp`) checks (12) directly. For fixed u, u', θ, z, it computes the original $M^2_{q\bar{q}}$ integral and the boundary term plus the correction integral with high-order composite Gauss–Legendre quadrature in k. The two agree to 10^{-7} relative, for both polarizations and for finite and large N_c.